## Theorem 4 (Fundamental Theorem of Arithmetic) (without proof)

"Every positive integer (except the number 1) can be represented in exactly one way apart from rearrangement as a product of one or more primes."

The fundamental theorem asserts that every composite number can be decomposed as a product of prime numbers and that the decomposition is unique. In the sense that there is one and only way to express the decomposition as product of primes.

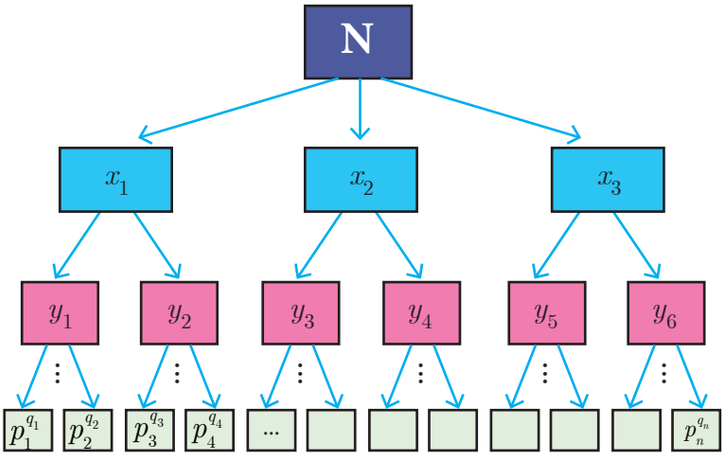


Fig.2.2

In general, we conclude that given a composite number N, we decompose it uniquely in the form $N = p_1^{q_1} \times p_2^{q_2} \times p_3^{q_3} \times \cdots \times p_n^{q_n}$ where $p_1, p_2, p_3, \ldots, p_n$ are primes and $q_1, q_2, q_3, \ldots, q_n$ are natural numbers.

First, we try to factorize N into its factors. If all the factors are themselves primes then we can stop. Otherwise, we try to further split the factors which are not prime. Continue the process till we get only prime numbers.

**Thinking Corner**

Is 1 a prime number?

### Illustration

For example, if we try to factorize 32760 we get

$$32760 = 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 7 \times 13$$

$$= 2^3 \times 3^2 \times 5^1 \times 7^1 \times 13^1$$

Thus, in whatever way we try to factorize 32760, we should finally get three 2's, two 3's, one 5, one 7 and one 13.

**Progress Check**

1. Every natural number except _____ can be expressed as _____.
2. In how many ways a composite number can be written as product of power of primes?
3. The number of divisors of any prime number is _____.

The fact that "Every composite number can be written uniquely as the product of power of primes" is called Fundamental Theorem of Arithmetic.

### 2.4.1 Significance of the Fundamental Theorem of Arithmetic

The fundamental theorem about natural numbers except 1, that we have stated above has several applications, both in Mathematics and in other fields. The theorem is vastly important in Mathematics, since it highlights the fact that prime numbers are the 'Building Blocks' for all the positive integers. Thus, prime numbers can be compared to atoms making up a molecule.



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